

Two minimizing log-polynomials in every even degree at least eight

Candidate counterexample for arXiv:2007.07952

autonomous proof-search run `fa_banach_001`

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Status. Candidate full negative answer, likely valid, pending expert review. For every even $d \geq 8$ we construct a normalized integrable log-concave function on $\mathbb{R}^2$ for which Problem 1 of the source paper has exactly two minimizers.

1 Source question

Alonso-Gutiérrez, González Merino and Villa consider, for even d , the problem of minimizing

$$\int_{\mathbb{R}^n} t \exp\{-g(x)^{1/d}\} dx = tn! |G_1(g)|, \quad G_1(g) = \{x : g(x) \leq 1\}, \quad (1)$$

over $t \geq 1$ and nonnegative homogeneous degree- d polynomials g , subject to

$$f(x) \leq t \exp\{-g(x)^{1/d}\}.$$

They prove existence for normalized integrable log-concave f . On PDF page 4, after Theorem 1.1, they note that uniqueness is known for $d = 2$ but is not known in general, and that they know no example with a nonunique minimization point [1].

We give such examples for every even $d \geq 8$.

2 Proof intuition

Write $f = e^{-u}$ and put $r = \log t$. The source reduction says that the fixed- r polynomial must contain

$$H_{e^r}(f) = \{u^* \leq r\}^\circ.$$

Thus we may design the sublevel bodies $D_r = \{u^* \leq r\}$ first and then recover f by convex duality.

We take a nested family D_r whose middle piece linearly interpolates, by Minkowski addition, between two scaled and rotated polynomial dual balls. At the endpoints, the optimal outer polynomial bodies are known exactly. Along the middle piece, symmetry forces every optimizer to have only the two Fourier modes 1 and $\cos(d\theta)$. Its containment constraint becomes the problem of finding the largest affine minorant of a concave power mean, hence the optimizer is exactly the chord joining the endpoint values.

The endpoint scale is selected so that their weighted volumes agree. A uniform Taylor calculation then gives

$$\log \frac{e^{r\lambda} v(r_\lambda)}{e^{r_0} v(r_0)} = \frac{2}{d} \left(1 - \frac{6}{d}\right) \lambda(1 - \lambda) \varepsilon^2 + O(\lambda(1 - \lambda) \varepsilon^3).$$

For $d \geq 8$ this is positive throughout $0 < \lambda < 1$ when ε is small. Simple homotheties make the objective strictly decreasing before the first endpoint and strictly increasing after the second. The endpoint pairs are therefore the two global minimizers.

In view of Lasserre's result, one might think in integrable functions $f : \mathbb{R}^n \rightarrow [0, +\infty)$ as the typical extension of compact sets to spaces of functions. Unfortunately, Problem 1 does not make sense in such a general case (see Example 3.1). Motivated by this fact, we solve the problem in the setting of log-concave integrable functions. Let $\mathcal{F}(\mathbb{R}^n)$ be the set of all log-concave integrable functions on $\mathbb{R}^n$.

Theorem 1.1. *Let $f \in \mathcal{F}(\mathbb{R}^n)$ with $\|f\|_\infty = f(0) = 1$ and $d \in \mathbb{N}$ even. Then there exists $(t_1, g_1) \in [1, +\infty) \times \mathbb{F}_d(\mathbb{R}^n)$ a solution of Problem 1.*

Notice that the problem considered in Theorem 1.1 was solved with uniqueness for $d = 2$ in [AGJV], in the even case, and in [LSW], in the general case, since for $g \in \mathbb{H}_2(\mathbb{R}^n)$ the set $G_1(g)$ is an ellipsoid provided that $|G_1(g)|$ is finite (see Lemma 2.3 (2)).

However, even this case is not solved with uniqueness in the proof of Theorem 1.1 with this point of view, since polarity does not work clearly between polynomials. More precisely, if we try to construct a proof by taking duals in the proof in [AGJV], we would need to take the polar of the ellipsoid $G_1(g)$, but the expression of the polynomial defining the polar ellipsoid in terms of g is not clear.

In the general case, the uniqueness is not straightforwardly obtained (although we do not know any example for which the minimization point is not unique). It seems to us that the proof would require some more convexity properties than the ones we have obtained.

The following similar problem is also posed. Unlike the case of Problem 1 above

Figure 1: The residual uniqueness question in the source paper (PDF page 4).

3 Reduction to a family of polar bodies

For a compact set $K \subset \mathbb{R}^2$ containing the origin, let

$$L_d(K) = \min \left\{ |G_1(g)| \mid \begin{array}{l} g \text{ is a nonnegative homogeneous degree-}d \text{ polynomial,} \\ K \subseteq G_1(g) \end{array} \right\}.$$

Lasserre's theorem gives a unique optimizing polynomial [2]; we use the theorem in precisely the form recorded in the source paper.

Lemma 1 (Realizing nested bodies). *Let $(D_r)_{r \geq 0}$ be compact origin-symmetric convex bodies satisfying*

1. $D_a \subseteq D_b$ for $a \leq b$;
2. $D_{(1-\lambda)a+\lambda b} \supseteq (1-\lambda)D_a + \lambda D_b$;
3. $D_0 = \{0\}$ and $\bigcup_{r \geq 0} D_r = \mathbb{R}^2$.

Then there is an even finite convex function w with $w(0) = 0$ and $\{w \leq r\} = D_r$. If $u = w^$ and $f = e^{-u}$, then f is even, log-concave, $f(0) = \|f\|_\infty = 1$, integrable, and*

$$\overline{H_{er}(f)} = D_r^\circ. \tag{2}$$

Consequently the best fixed- r value in (1), with the fixed factor $2!$ omitted, is

$$\Phi(r) = e^r L_d(D_r^\circ). \tag{3}$$

Proof. Set $w(x) = \inf\{r : x \in D_r\}$. The nesting gives $\{w \leq r\} = D_r$ (closedness also handles the infimum), and the Minkowski concavity gives convexity of w . The remaining assertions except integrability follow from elementary Fenchel duality. In particular $u^* = w$, and the polarity calculation in Lemma 3.3 of [1] gives (2).

In our application below, $D_r = (r/r_0)D_+$ for $0 \leq r \leq r_0$. Thus $w(x) = r_0\|x\|_{D_+}$ wherever $w(x) \leq r_0$, and

$$u(y) = \sup_x (\langle x, y \rangle - w(x)) \geq \sup_{0 \leq a \leq 1} a(h_{D_+}(y) - r_0) = (h_{D_+}(y) - r_0)_+.$$

Since D_+ contains a Euclidean ball, $h_{D_+}(y) \geq c|y|$ for some $c > 0$. Hence $e^{-u(y)} \leq e^{-(c|y|-r_0)_+}$ is integrable. The equivalence between feasibility and $H_{e^r}(f) \subseteq G_1(g)$ proved in Lemma 3.1 of [1] now yields (3). $\square$

4 The construction and the exact fixed-height optimizer

Fix an even integer $d \geq 8$, write $\alpha = 1/d$, and fix $\kappa > 0$. For a sufficiently small $\varepsilon \in (0, 1)$, use polar coordinates $x = \rho(\cos \theta, \sin \theta)$ and define

$$p_+(x)^d = \rho^d(1 + \varepsilon \cos d\theta), \quad p_-(x)^d = s^d \rho^d(1 - \varepsilon \cos d\theta), \quad \sigma = \left(\frac{1 + \varepsilon}{1 - \varepsilon}\right)^\alpha, \quad s = \sigma e^{\kappa \varepsilon^2}.$$

The expressions on the right are homogeneous polynomials: they equal $(x^2 + y^2)^{d/2} \pm \varepsilon \operatorname{Re}(x + iy)^d$, with the second one multiplied by s^d . For small ε , $p_\pm$ are norms. Indeed the unit circle of p_+ has radial function $R(\theta) = (1 + \varepsilon \cos d\theta)^{-1/d}$, and $R^2 + 2(R')^2 - RR'' = 1 + O_d(\varepsilon)$ uniformly; the same applies to p_- . Let $C_\pm = \{p_\pm \leq 1\}$ and $D_\pm = C_\pm^\circ$.

The ratio p_-/p_+ ranges from $s/\sigma = e^{\kappa \varepsilon^2}$ to $s\sigma$. Thus $C_- \subset \operatorname{int} C_+$ and, after polarity, $D_+ \subset \operatorname{int} D_-$. Put

$$\Delta = 2 \log s, \quad r_0 = \frac{1}{2}, \quad r_1 = r_0 + \Delta, \quad \eta = \frac{(s/\sigma) - 1}{2\Delta(s/\sigma)}.$$

and define, for $r \geq 0$,

$$D_r = \begin{cases} \frac{r}{r_0}D_+, & 0 \leq r \leq r_0, \\ (1 - \lambda)D_+ + \lambda D_-, & r = r_0 + \lambda\Delta, \quad 0 \leq \lambda \leq 1, \\ (1 + \eta(r - r_1))D_-, & r \geq r_1. \end{cases} \quad (4)$$

This family is increasing. Its support function is concave in r : it is linear on each of the three intervals, and at the two knots its slopes obey

$$\frac{h_{D_-} - h_{D_+}}{\Delta} \leq \frac{h_{D_+}}{r_0}, \quad \frac{h_{D_-} - h_{D_+}}{\Delta} \geq \eta h_{D_-}.$$

Indeed, on writing $q = h_{D_-}/h_{D_+} = p_-/p_+$, we have $s/\sigma \leq q \leq s\sigma$. The first assertion follows for small ε from

$$\frac{s\sigma - 1}{\Delta} \longrightarrow 1 < \frac{1}{r_0} = 2,$$

and the second follows directly from $(q - 1)/(\Delta q) \geq ((s/\sigma) - 1)/(\Delta(s/\sigma)) = 2\eta$. Moreover $\eta > 0$, so the family exhausts $\mathbb{R}^2$, and $\eta \rightarrow 0$ as $\varepsilon \rightarrow 0$. Hence Lemma 1 applies.

We next compute $L_d(D_r^\circ)$ on the middle interval. Since the support function of $D_\pm = C_\pm^\circ$ is $p_\pm$, the polar of the Minkowski sum in (4) has radial function

$$R_\lambda(\theta) = \frac{1}{(1-\lambda)(1+\varepsilon \cos d\theta)^\alpha + \lambda s(1-\varepsilon \cos d\theta)^\alpha}. \quad (5)$$

Lemma 2 (Exact optimizer on the middle segment). *For $0 \leq \lambda \leq 1$, set*

$$\begin{aligned} a_\lambda &= (1-\lambda)(1+\varepsilon)^\alpha + \lambda s(1-\varepsilon)^\alpha, \\ b_\lambda &= (1-\lambda)(1-\varepsilon)^\alpha + \lambda s(1+\varepsilon)^\alpha, \\ A_\lambda &= \frac{a_\lambda^d + b_\lambda^d}{2}, \quad B_\lambda = \frac{a_\lambda^d - b_\lambda^d}{2}. \end{aligned}$$

Then the unique Lasserre polynomial of $D_{r_0+\lambda\Delta}^\circ$ is

$$g_\lambda(\rho, \theta) = \rho^d(A_\lambda + B_\lambda \cos d\theta). \quad (6)$$

In particular

$$L_d(D_{r_0+\lambda\Delta}^\circ) = \frac{1}{2} \int_0^{2\pi} (A_\lambda + B_\lambda \cos d\theta)^{-2/d} d\theta. \quad (7)$$

Proof. The body in (5) is invariant under reflection and rotation by $2\pi/d$. Uniqueness of the Lasserre polynomial makes the optimizer invariant under these symmetries. A real homogeneous degree- d polynomial restricted to the circle has Fourier frequencies $d, d-2, \dots, -d$. Rotation by $2\pi/d$ retains only $0, \pm d$, and reflection removes the sine term; hence it has the form $\rho^d(A + B \cos d\theta)$.

Containment is equivalent, with $z = \cos d\theta$, to

$$A + Bz \leq F_\lambda(z) := ((1-\lambda)(1+\varepsilon z)^\alpha + \lambda s(1-\varepsilon z)^\alpha)^d, \quad -1 \leq z \leq 1. \quad (8)$$

The weighted power mean $M_\alpha(x, y) = ((1-\lambda)x^\alpha + \lambda y^\alpha)^{1/\alpha}$ is concave and coordinatewise increasing for $0 < \alpha \leq 1$ (the reverse Minkowski inequality). Since $x = 1 + \varepsilon z$ and $y = s^d(1 - \varepsilon z)$ are affine in z , F_λ is concave. Therefore its endpoint chord

$$\ell_\lambda(z) = \frac{F_\lambda(1) + F_\lambda(-1)}{2} + \frac{F_\lambda(1) - F_\lambda(-1)}{2}z = A_\lambda + B_\lambda z$$

is feasible. Every other feasible affine function $q(z)$ obeys $q(\pm 1) \leq \ell_\lambda(\pm 1)$, so the affine difference $q - \ell_\lambda$ is nonpositive throughout $[-1, 1]$. Thus its polynomial sublevel body contains that of (6), proving optimality. Formula (7) is the polar-coordinate area formula. $\square$

5 The two minima

Lemma 3 (Middle barrier). *For every fixed even $d \geq 8$, all sufficiently small $\varepsilon > 0$ satisfy*

$$e^{r_0+\lambda\Delta} L_d(D_{r_0+\lambda\Delta}^\circ) > e^{r_0} L_d(C_+) \quad (0 < \lambda < 1),$$

whereas equality holds for $\lambda = 0, 1$.

Proof. Let V_λ denote the right-hand side of (7) and define

$$Q(\lambda, \varepsilon) = \lambda\Delta + \log V_\lambda - \log V_0.$$

At $\lambda = 0$, $g_\lambda = p_+^d$; at $\lambda = 1$, $g_\lambda = p_-^d$. Rotation does not change area, so $V_1 = s^{-2}V_0$ and $Q(0, \varepsilon) = Q(1, \varepsilon) = 0$ exactly.

All quantities are real analytic near the compact square $[0, 1] \times \{0\}$. Direct Taylor expansion, using

$$\frac{1}{2\pi} \int_0^{2\pi} \cos(d\theta) d\theta = 0, \quad \frac{1}{2\pi} \int_0^{2\pi} \cos^2(d\theta) d\theta = \frac{1}{2},$$

gives, uniformly for $0 \leq \lambda \leq 1$,

$$Q(\lambda, \varepsilon) = 2\alpha(1 - 6\alpha)\lambda(1 - \lambda)\varepsilon^2 + O(\lambda(1 - \lambda)\varepsilon^3). \quad (9)$$

For completeness, here is an auditable coefficient calculation. Uniformly in λ ,

$$\begin{aligned} A_\lambda &= 1 + 2\lambda\varepsilon + a_2(\lambda)\varepsilon^2 + O(\varepsilon^3), \\ B_\lambda &= (1 - 2\lambda)\varepsilon + b_2(\lambda)\varepsilon^2 + O(\varepsilon^3), \\ a_2(\lambda) &= \lambda \left(4\alpha(1 - \lambda) + 4\lambda - 2 + \frac{\kappa}{\alpha} \right), \\ b_2(\lambda) &= 2\lambda(2\alpha\lambda - 2\alpha - 2\lambda + 1), \\ \Delta &= 4\alpha\varepsilon + 2\kappa\varepsilon^2 + O(\varepsilon^3). \end{aligned}$$

Writing $c_\lambda = B_\lambda/A_\lambda$, the angular average in (7) gives

$$\log V_\lambda = \log \pi - 2\alpha \log A_\lambda + \frac{\alpha(2\alpha + 1)}{2} c_\lambda^2 + O(\varepsilon^3).$$

Substitution and subtraction of the value at $\lambda = 0$ cancel both the linear term and the κ term, leaving

$$2\alpha(1 - 6\alpha)\lambda(1 - \lambda).$$

The displayed remainder has the endpoint factors because Q is analytic and vanishes identically at $\lambda = 0, 1$: successive analytic division by λ and $1 - \lambda$ makes the quotient bounded on a fixed neighborhood. Since $1 - 6\alpha > 0$ for $d \geq 8$, (9) is positive for every $0 < \lambda < 1$ once ε is sufficiently small. $\square$

Theorem 1 (Negative answer to the uniqueness question). *For every even integer $d \geq 8$ there is an even normalized integrable log-concave function $f : \mathbb{R}^2 \rightarrow [0, 1]$ for which Problem 1 of [1] has exactly two minimizing pairs.*

Proof. Choose ε as in Lemma 3, small enough also that $\eta < 1/2$, construct D_r by (4), and let f be supplied by Lemma 1. On $0 < r \leq r_0$, homogeneity gives

$$L_d(D_r^\circ) = L_d((r_0/r)C_+) = (r_0/r)^2 L_d(C_+),$$

so $\Phi(r)$ is strictly decreasing there because $r \leq r_0 = 1/2 < 2$. On $r \geq r_1$,

$$L_d(D_r^\circ) = (1 + \eta(r - r_1))^{-2} L_d(C_-),$$

and

$$\frac{d}{dr} \log \Phi(r) = 1 - \frac{2\eta}{1 + \eta(r - r_1)} > 0.$$

The endpoint separation is $\Delta = 2 \log s$, so $e^{r_1}|C_-| = e^{r_0 + \Delta} s^{-2} |C_+| = e^{r_0} |C_+|$. Here $L_d(C_\pm) = |C_\pm|$, because $C_\pm$ are themselves the sublevel bodies of $p_\pm^d$, while no containing body can have smaller volume. Lemma 3 puts every point of the middle segment strictly above this common value. The two outer pieces move away from the common value, so the only global minimizers occur at $r = r_0$ with polynomial p_+^d and at $r = r_1$ with polynomial p_-^d . They are distinct because $\varepsilon > 0$. $\square$

6 Verification and limitations

The accompanying script independently checks the symbolic coefficient in (9) and samples the exact integral (7). Those checks are sanity tests, not proof. The argument proves nonuniqueness for every even $d \geq 8$ in dimension two. It does not decide the borderline degrees $d = 4$ and $d = 6$: the quadratic coefficient is negative for $d = 4$ and zero for $d = 6$.

A bounded novelty search on 13 August 2026 covered the run indexes, exact title/id, author/title combinations, and arXiv/web searches for log-polynomial, d -Lasserre–Löwner, uniqueness, and nonunique minimizers. It found no later resolution. This is not proof of novelty.

Human-review recommendation. Check (i) realization of the nested family, (ii) the support-slope bookkeeping at the two knots, (iii) the invariant-polynomial reduction, (iv) concavity and chord optimality in Lemma 2, and (v) the uniform endpoint factor in (9).

References

- [1] D. Alonso-Gutiérrez, B. González Merino, and R. Villa, *Best approximation of functions by log-polynomials*, J. Funct. Anal. 280 (2021), 108999; arXiv:2007.07952.
- [2] J.-B. Lasserre, *A generalization of Löwner–John’s ellipsoid theorem*, Math. Program. 152 (2015), 559–591; arXiv:1302.1056.